NEWTON CLASSES

Class 10 Science — Quick Revision Notes

Starter revision resource • Verify the current Bihar Board syllabus with your school.

Chemical Reactions

• A balanced chemical equation has equal numbers of each type of atom on both sides.

• Combination: two or more reactants form one product. Decomposition: one compound breaks into simpler substances.

• Oxidation commonly means addition of oxygen or removal of hydrogen; reduction is the reverse in many school-level examples.

Acids, Bases and Salts

• Acids produce H■/H■O■ ions in aqueous solution; bases produce OH■ ions in aqueous solution.

• pH below 7 is acidic, 7 is neutral at 25°C, and above 7 is basic.

• Always add acid to water slowly, not water to concentrated acid.

Metals and Non-metals

• Metals generally conduct heat and electricity and are malleable; exceptions exist.

• More reactive metals can displace less reactive metals from their salt solutions.

• Corrosion can be reduced by painting, oiling, galvanising or alloying, depending on the material.

Life Processes

• Nutrition, respiration, transportation and excretion are major life processes.

• Photosynthesis uses carbon dioxide and water to produce glucose and oxygen in the presence of light and chlorophyll.

• In humans, the heart pumps blood; alveoli in lungs provide a large surface for gas exchange.

Control and Coordination

• Neurons carry electrical impulses; synapses pass signals between neurons using chemical messengers.

• Plant hormones help regulate growth and responses; auxin is involved in phototropism.

• Hormones are chemical messengers released by endocrine glands.

Reproduction and Heredity

• Asexual reproduction generally involves one parent; sexual reproduction involves fusion of gametes.

• Genes are segments of DNA that influence inherited traits.

• In humans, the egg contributes an X chromosome; sperm contributes X or Y, usually determining chromosomal sex.

Light and Electricity

• Mirror formula: 1/f = 1/v + 1/u using the Cartesian sign convention.

• Lens formula: 1/f = 1/v – 1/u; magnification for a lens m = v/u.

• Ohm’s law: V = IR. Electric power: P = VI = I^2R = V^2/R.

• Series resistors: R = R■ + R■ + …; parallel: 1/R = 1/R■ + 1/R■ + … .

Environment

• Food chains transfer energy between trophic levels; energy available generally decreases at higher levels.

• Biodegradable substances can be broken down by microorganisms; non-biodegradable waste can persist longer.

• Ozone in the stratosphere absorbs much of the Sun’s harmful ultraviolet radiation.

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